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Energy-efficient spintronic based neuromorphic computing system using current mode track and termination circuit

Pegah Shafaghi, Mehdi Dolatshahi and Hooman Farkhani, *Member, IEEE*

Abstract— Spintronic nano-devices have shown great potential to reduce the energy consumption of neuromorphic computing systems (NCSs). In the spintronic-based NCSs, the switching or oscillation of a magnetic tunnel junction (MTJ) is a common approach to mimic neuron firing. However, still there is a gap between the performance (operation/sec/Watt/cm²) of the human brain and NCSs. To mitigate this gap, it is essential to further decrease the energy consumption and the delay of the NCS. The high energy consumption of the MTJ-based NCS is mostly related to the high current needed to switch the MTJ state. Hence, some previous methods tried to perform real-time tracking of the MTJ state by monitoring the voltage across the MTJ and cut-off its current immediately after switching. However, due to the small voltage changes after switching, these methods suffer from high-power consumption. In this paper, a new method based on the tracking of the MTJ current (instead of its voltage) and terminating the MTJ current after switching is proposed. Due to the large changes in the MTJ current after switching (about 40%), there is no need to use an amplifier in the proposed common-mode tracking and terminating circuit (CM-TTC). The simulation results in 65-nm CMOS technology confirm that the proposed CM-TTC technique can improve the energy consumption and speed of a typical NCS by 53% and 2X. Moreover, the power consumption, delay and area overhead of CM-TTC is reduced by 12.8%, 73% and 95% compared with the best state-of-the-art track and termination circuits.

Index Terms—Energy-efficiency, Memristor, Magnetic tunnel junction, Neuromorphic computing system, Spintronics.

I. INTRODUCTION

Artificial intelligence (AI) has witnessed explosive growth over the past two decades. The core processing unit of AI is an artificial neural network (ANN) inspired by the human brain with remarkable accuracy progress in cognitive applications [1]. Despite all the progress in ANN algorithms, the biggest challenge of AI is the low performance (operation/Joul/area) of their hardware implementation using typical computers (CPUs and GPUs). Brain-inspired hardware implementation of ANNs algorithms, called neuromorphic computing systems (NCSs), has shown to be a promising solution to deal with the low performance of implementing AI

on typical computers (e.g. TrueNorth by IBM [2] and Loihi by Intel [3]). NCSs overcome the limitation of the word-at-a-time thinking of typical computers by doing massively parallel data processing, similar to the brain. Despite the higher density and energy efficiency of implementing ANN algorithms on NCSs, they are still far from human brain performance.

There are different state-of-the-art implementations of NCS using CMOS (digital and analog) and emerging technologies (e.g. spintronic, phase change memory and memristors). Implementation of NCSs using conventional digital and analog CMOS is area-and energy-inefficient [4]. This led to a significant effort to examine the hybrid implementation of NCS through implementing the neural networks using emerging technologies and CMOS circuits as controlling blocks. Thanks to the advancement of nanotechnology, one of the most promising hybrid implementations of NCS is the combination of nanoscale spin-based components and CMOS devices (i.e. spintronics). Spintronics has made it possible to implement NCSs with higher density and lower power consumption [5-6]. Hardware implementation of the NCS requires massively parallel components (act as neurons) interacting with each other and the external world through the adaptive or programmable devices acting as synapses. In spintronic-based NCS, neurons and synapses would be implemented using magnetic elements while CMOS circuits can be used as input interfacing circuits and readout circuits for sensing outputs. The design of neurons determines the performance and the integration density of the NCS regardless of the network topology [7-10].

One of the most popular spintronic-based implementations of NCS is to use MTJs and memristors for mimicking the neuron and synapse functionalities, respectively. However, the MTJ switching behavior is innately stochastic. Hence, there is a need for passing the current to MTJ for a time period longer than the worst-case switching time of MTJ to guarantee its correct switching when its current is higher than the threshold current. This means in most cases, the switching current will flow through the MTJ even after its switching. This leads to high energy consumption during the MTJ switching process. In the literature, attempts have been made to solve this problem using tracking the voltage across the MTJ for detecting its switching and terminating its current [11-13]. However, due to the low voltage change after switching, those methods need a power-hungry amplifier which results in a high-power

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consumption. Note that MTJ switching consumes the majority of the energy consumption in spintronic-based NCS [14].

In this paper, a new technique based on tracking the MTJ current (instead of its voltage) and terminating its current immediately after switching is proposed. Considering the high changes in the MTJ current after switching (about 40%), there is no need to use any amplifier in the proposed circuit, which reduces the power consumption of the proposed circuit in comparison with the state-of-the-art.

The rest of the paper is organized as follows: Section II briefly discusses the implementation of MTJ-based neurons. In section III, the proposed CM-TTC method is discussed. Section IV presents the simulation results and comparisons. Finally, section V concludes the paper.

II. STATE OF THE ART

A. Performance of the Neuron

In a biological neural network, neurons which are known as a set of parallel processors work together simultaneously and synapses are in charge of transferring signals among different neurons. When different input signals come from neurons in the previous layer (pre-neurons), the post-neuron integrates these signals in the form of membrane potential. If the membrane potential of the post-neuron reaches a threshold potential, it produces a wave-shaped signal (called an action potential) as shown in Fig.1. Then, the synapse transmits the action potential to the next neuron. However, it is not a simple transition but the synapse amplifies or weakens the signal along the path based on past events.

The artificial neural networks are modeled based on the biological neural networks, in which proper modeling of the neuron is the starting point. Each neuron has a single output and a set of inputs, as shown in Fig. 2. The inputs are weighted and as mentioned above the sum of these weighted inputs is applied to the activation function. There are different types of activation functions discussed in the literature [15-16]. However, in the MTJ implementation of a neuron based on switching, the step function will be used. The activation function produces the output based on the summation of the input signals. The inputs, named as $v_1, v_2, \dots, v_n$, are weighted by the coefficients $w_1, w_2, \dots, w_n$ and are applied to the input of the neuron. These weights

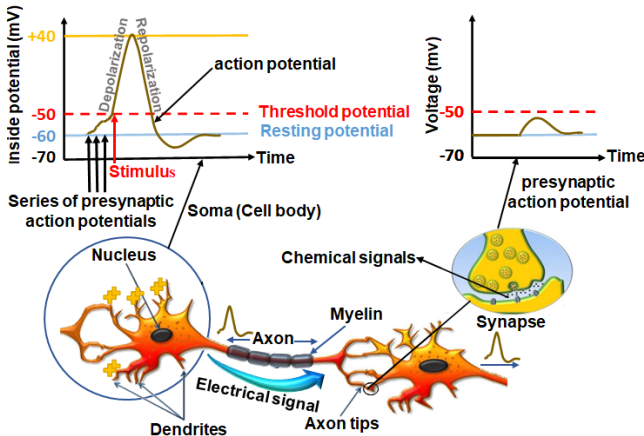


Fig. 1. The simplified model of human brain computation.

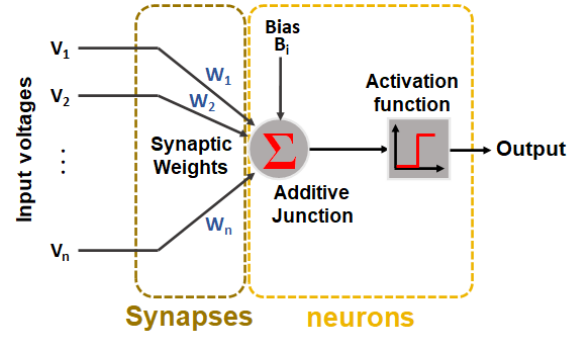


Fig. 2. The typical model of an artificial neuron.

represent the biological synaptic functionalities in the neurons [17]. The operation of the step function can be given as:

$$Output = \begin{cases} 1 & \text{if } \sum W_i V_i + B_i \geq Threshold \\ 0 & \text{if } \sum W_i V_i + B_i < Threshold \end{cases} \quad (1)$$

where W_i is the corresponding synapse weight, B_i is the neuron bias and V_i indicates the neuron input.

B. Magnetic Tunnel Junction basis

MTJ is one of the most promising spin-based devices. Fig. 3 (a) shows the basic structure of an MTJ. It consists of two ferromagnetic layers with a non-conductive thin tunneling barrier that is sandwiched between the two ferromagnetic layers. The pinned layer (PL), also known as a fixed layer, is a thick ferromagnetic layer with a fixed magnetic orientation. In contrast to the PL, the free layer (FL) magnetization direction can be changed by applying an external magnetic field and/or passing current in the right direction [18]. As shown in Fig. 3 (a), the magnetization direction of the FL relative to the PL determines the MTJ resistive behavior. When the magnetization directions of two layers are in parallel (P-state), the MTJ resistance (R_P) is low, however, if the magnetization directions of the two magnetic layers are anti-parallel (AP-state), the MTJ resistance (R_{AP}) is high. Fig. 3 (b) shows the house curve characteristic of the MTJ while switching from AP to P and vice versa. As illustrated in Fig. 3 (b), the MTJ switching behavior is asymmetric, i.e. the MTJ switching time and energy consumption while switching from P to AP and AP to P are different. In most of the MTJ implementations, the P to AP state transition has a higher switching threshold current. It is because the PL weakens the spin electrons [19-21]. In other words, the PL magnetic field prevents the FL spins from switching to the opposite direction of the PL magnetic direction [22]. This prevention leads to an increased current needed for the MTJ

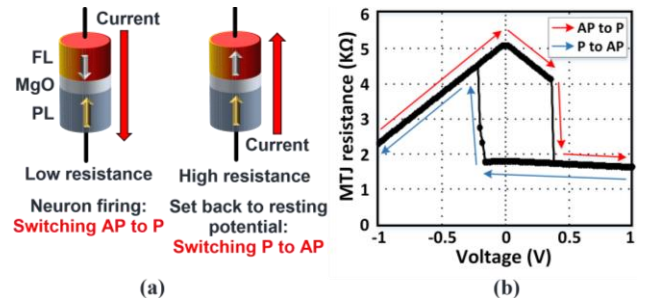


Fig. 3. (a) The schematic of the MTJ and the switching behavior of the MTJ. (b) The house curve characteristic of the MTJ.

switching. In contrast, while switching from AP to P, PL magnetic field will ease the switching process, leading to a lower threshold current. As a result, the initial state of the MTJ-based neuron will be set at AP-state to reduce the power consumption and delay of the NCS [23]. If a sufficient current flows from FL to PL, the MTJ state switches to P-state and the MTJ resistance will decrease to R_p [24-26]. Accordingly, by using a readout circuit, it is possible to evaluate the resistance of MTJ and determine whether the neuron has fired or not.

C. MTJ and Memristor Functionality in Neuromorphic Computing System

The human brain is a massively parallel computing system that employs the neurons as simple computing units and synapses as adjustable weighted connections between neurons as described in section II. The NCS, tries to mimic the way that the brain does computing. To implement the NCS, it is needed to implement neuronal and synaptic functionalities through electronic elements. Hence, memristors can be used to perform synaptic functionality to get low-power operation [27-28]. The memristor resistance can be tuned by electrical signals. Each input is connected to the neuron by the memristor. Hence, different inputs and their corresponding memristors apply different currents to the post-neuron. These currents will be added together and the total current passes through the MTJ which is used to mimic the neuron. If the total current will be higher than the critical current ($I_{sum} > I_{critical}$), MTJ will switch. Then, the MTJ switching will be detected by the readout circuit as neuron firing. However, the relatively high threshold current of switching and the stochastic nature of the MTJ switching increase the energy consumption of the MTJ-based NCS. To this end, a novel CM-TTC technique is proposed to mitigate the negative impact of the MTJ switching stochasticity.

III. PROPOSED TRACKING AND TERMINATING TECHNIQUE

In this section, the structure of the proposed CM-TTC will be explained. As mentioned above, the MTJ is used to mimic neuron functionality, however, the high energy consumption needed for the MTJ switching increases the energy consumption of the NCS. The CM-TTC tries to decrease the energy consumption of the MTJ switching using high changes in the MTJ current after switching and cutting off the MTJ current, immediately.

A. Main Idea

The schematic of the CM-TTC-based NCS is shown in Fig. 4 (a). The proposed CM-TTC technique can be added to a typical NCS to decrease its energy consumption. A significant part of the energy consumption of the MTJ-based NCS is related to the high current required to switch the MTJ state. Hence, the CM-TTC is proposed to decrease the total energy consumption by tracking and terminating the MTJ current immediately after switching. To sense the MTJ switching, CM-TTC tracks the MTJ current changes. Due to the high changes in the MTJ current after switching (about 40%), there is no need to use an amplifier in the CM-TTC. This will decrease the energy consumption and delay of the CM-TTC-based NCS, significantly. In the literature, different methods have been

proposed to lower the energy consumption of MTJ switching. The real-time sensing (RTS) circuit proposed in [11] and the Self-Referenced Differential Write Termination (SDWT) circuit [13] track the MTJ state during the stimulation phase simultaneously by sensing its voltage and cutting off the MTJ current immediately after the MTJ switching. Despite the decrease in the energy consumption of the MTJ using the RTS and SDWT circuits, they consume a remarkable amount of energy due to the use of an amplifier. The amplifier is needed to amplify the voltage changes after switching (about 40mv) to be comparable with a reference voltage using a comparator. In the RTS-based NCS, 25-40% of total energy consumption is related to the RTS circuit [11]. It is because the amplifier in the RTS circuit is turned on for 90% of the stimulation phase. Moreover, the delay of the RTS circuit is about 0.55ns and during this time not only the RTS consumes energy, but also a high current passes through the MTJ. Fig. 4 (b) shows the circuit implementation of the CM-TTC. It consists of five main blocks including a current mirror, a current comparator, a rail-to-rail output buffer, two OR logic gates and a rise edge detector. The idea is to compare the MTJ current with a reference current equal to the average current of the MTJ after and before switching. To do that the MTJ current is mirrored to the current comparator via M_1 and M_2 . Then, the current comparator formed by M_{2-8} transistors compares the MTJ current with a reference current ($(I_{MTJ-P} + I_{MTJ-AP})/2$). The reference current, formed by M_8 , is mirrored to the M_6 through M_7 transistor. The size of M_8 is designed in a way that the reference current will be the average of I_{M5} before and after switching. On the other hand, the gate of M_8 is connected to the common node between the transmission gate (TG1) and M_1 . The main advantage of this biasing method is that it will help the CM-TTC circuit to detect switching more effectively. Before switching, the MTJ resistance is high (R_{AP}) and the gate voltage of M_8 is relatively low. As a result, the reference current is high. After switching, because the gate voltage of M_8 will increase, the reference current will decrease. This will act as positive feedback and increase the voltage of the COM node even more which leads to a faster current cut-off in the MTJ branch. Finally, the current of the M_5 transistor (MTJ current) is compared with the reference current passing through M_6 ($(I_{MTJ-P} + I_{MTJ-AP})/2$). Before switching, MTJ is in AP-state and its current is less than the reference current. Hence, I_{M5} is smaller than I_{M6} which will push M_6 to the linear region. This will reduce the voltage of the COM node to voltages below $V_{DD}/2$. Then, this voltage will be inverted and amplified to V_{DD} at BU which will keep TG1 (M_{ST}) turned on (off) through the OR gates. The rise edge detector (RED) aims to detect the positive edges of Pulse1 and create a short pulse immediately afterward. This short pulse will keep TG1 on through the OR gates when Pulse1 will be switched from '0' to '1'. Without RED block, both Pulse1 and $\overline{Pulse1}$ will be '0' for a short time during the rising edge of Pulse1 which will turn off TG1 irreversibly. This means that the TG1 will remain off even after Pulse1 switching is complete. As soon as the MTJ will be switched to P-state, its current will increase to I_{MTJ-P} due to the MTJ resistance reduction. Hence, I_{M5} will be more than I_{M6} that will push M_5 to the linear region and increase the voltage of the COM node to voltages above $V_{DD}/2$. Again, this voltage will be

TABLE I
NCS PARAMETERS

MTJ [30]	Free Layer Dimensions	40nm×116nm×1.5nm
	Oxide Thickness (T_{MgO})	1.15 nm
	Saturation Magnetization (M_s)	850 emu/cm ³
	Damping Factor (α)	0.028
CMOS Transistor [29]	Technology	65 nm
	V_{DD}	1V
Memristor [31]	R_{ON}/R_{OFF}	10/10000
	Thin-film thickness	3 nm
	K_{ON}	-8×10^{-13}
	K_{OFF}	8×10^{-13}
	α	3

simulation in the HSPICE simulator are considered as 1V and 25°C, respectively. The compact modular MTJ model [30] which is working based on a stochastic LLG solver block is used. For the memristor, the TEAM model [31] is used in simulations due to its flexibility and accuracy. Table I shows the typical values of MTJs, memristors and CMOS transistors used in the proposed CM-TTC-based NCS.

A. Effect of Process Variation on the MTJ Switching Time

The process variations (PVs) of MTJ, transistors and memristors can change the switching time (ST) of the MTJ. Hence, in order to investigate the performance of the proposed CM-TTC in the presence of PVs, Monte Carlo simulations of 1000 iterations are run for different MTJ current values. For each MTJ current, the mean value and standard deviation of the ST are calculated. Fig. 6 shows the 6-sigma variations of ST for different MTJ current values. The shortest ST is achieved as 1.87ns at $I_{MTJ}=170\mu A$ and the longest ST is calculated as 13.27ns at $I_{MTJ}=120\mu A$. Fig. 6 shows the ST histogram and its fitted positively skewed function for $I_{MTJ}=120\mu A$. Both thermal fluctuations and process variations effects will affect the standard deviation. However, the thermal fluctuations are the main source of the ST mean value reduction [32].

B. Transient Simulation

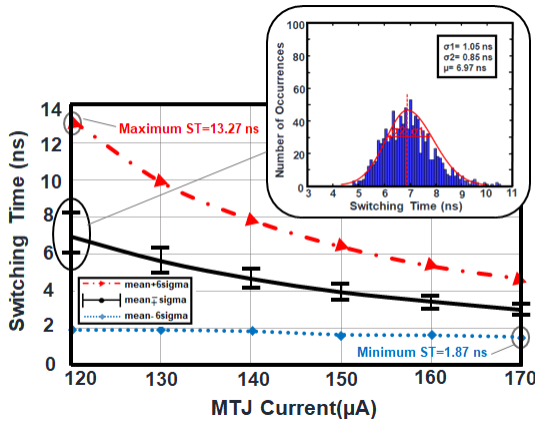


Fig. 6. The standard deviation and mean value of the MTJ switching time for various MTJ current values. MC is performed for each current with 1000 iterations. The 6-sigma variations for different currents of MTJ. The maximum and minimum ST of MTJ are calculated as 13.27ns at $I_{MTJ}=120\mu A$ and 1.87ns at $I_{MTJ}=170\mu A$. The histogram for the switching time of MTJ and the Gaussian function fitted to the histogram by running MC simulation with 1000 iterations for the NCS is depicted.

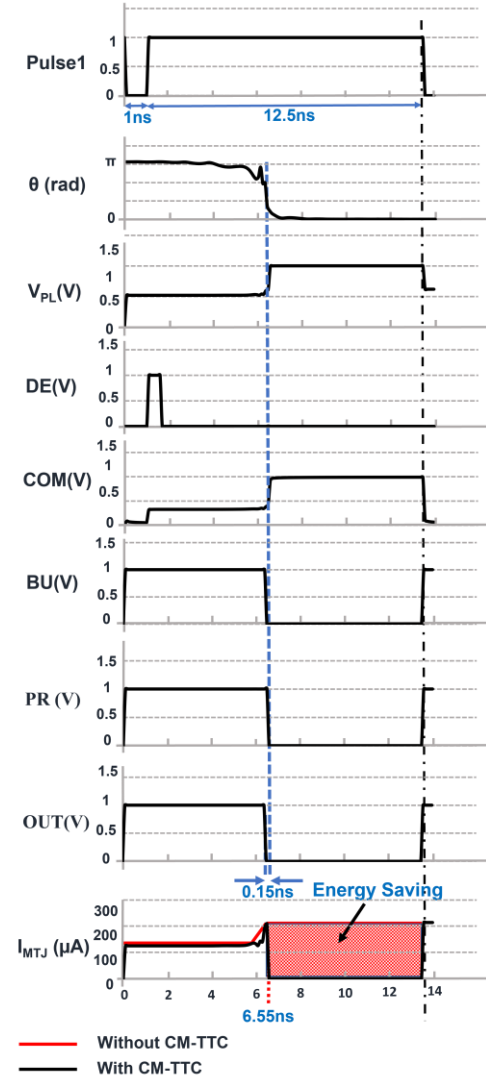


Fig. 7. Transient simulation of the NCS with proposed tracking and terminating technique

Fig. 7 shows the transient simulation of the proposed CM-TTC circuit. Transient simulation is done for $V_{DD}=1V$. The angle (θ), between the magnetization of the two ferromagnetic layers of MTJ (PL and FL), determines the MTJ state. If $\theta=0$, it means that the MTJ is in the P-state, while $\theta=\pi$ means that the MTJ is in the AP-state. The simulation of NCS starts at $t=1ns$ by enabling the Pulse1 signal. The MTJ is switched at 6.97ns after enabling Pulse1. For the ST of 6.97ns, the high skew and low skew standard deviations are calculated as 1.05ns and 0.85ns, respectively, using a Monte Carlo (MC) simulation with 1000 iterations (the details of the MC simulation can be found in the next subsection). Considering the high skew standard deviation of 1.05ns, the 6sigma+mean value of ST is calculated as 13.27ns. Moreover, the low skew standard deviation determines the 6sigma-mean value of ST as 1.87ns. Hence, the fastest ST and the slowest ST in the presence of 6sigma variations are 13.27ns and 1.87ns, respectively. As a result, to guarantee the MTJ switching in the typical NCS in the presence of 6sigma variations, a pulse period longer than 13.27ns is required. Because the MTJ switching never happen before 1ns (the fastest ST is calculated as 1.87ns), the CM-TTC circuit will be turned on with 1ns delay with respect to Pulse1 to reduce the

total power consumption of the CM-TTC-based NCS. When the CM-TTC circuit is on ($t > 1\text{ns}$), the MTJ current will be copied to the branch of the current comparator and will be compared with the reference current ($(I_{\text{MTJ-P}} + I_{\text{MTJ-AP}})/2$). Before the MTJ switching, MTJ is in AP-state and its current is lower than the reference voltage. Hence, the output voltage of the current comparator (COM) is less than $V_{\text{DD}}/2$ that will be inverted and amplified to 1V at the BU. This will keep TG1 turned on through OR gates. When the MTJ switching will happen ($t = 6.97\text{ns}$), the MTJ current will jump above the reference current. As a result, the output voltage of the current comparator (V_{COM}) will go above $V_{\text{DD}}/2$. This will force the voltage of the BU to be switched from '1' to '0'. This will cause the gate voltage of the control transistor (TG1) will be switched to '0' through the OR gates that will turn TG1 off and cut off the MTJ current immediately after switching instead of wasting energy and time until 13.5ns. The delay of the CM-TTC circuit is estimated to be 0.15ns as shown in Fig. 7.

C. Energy Consumption Reduction in CM-TTC

To calculate the total energy consumption of the CM-TTC-based NCS, the energy consumption of the MTJ branch (memristors, MTJ, M1 and transmission gate) and the CM-TTC have been calculated. The energy consumption of the CM-TTC-based NCS is calculated as 1088fJ at $V_{\text{DD}} = 1\text{V}$ and $I_{\text{MTJ}} = 120\mu\text{A}$. In the typical NCS, as long as the Pulse1 signal is active, the current flows in the memristor, MTJ and transistors. Hence, the energy consumed by the typical NCS is $\sim 2316\text{fJ}$.

When Pulse1 rises to '1', the MTJ is stimulated and the CM-TTC senses the MTJ state, simultaneously. In case, the MTJ switching occurs, the CM-TTC cuts off the current path by turning off the TG1 and stimulates the post-neuron. This reduces the energy consumption of NCS from 2316fJ to 1088fJ. The energy consumption of the CM-TTC based NCS at the optimal delay (calculated as 0.15ns in section D) is achieved as 1088fJ. Hence, the energy consumption of CM-TTC-based NCS is reduced by 53% compared to the typical NCS.

The total energy consumption of the CM-TTC-based NCS and the typical NCS are shown in Fig. 8 for different MTJ currents. In the typical NCS, after MTJ switching, the MTJ current will flow until the end of stimulation (13.5ns). By increasing the MTJ current, both the average and worst-case ST

will reduce at the cost of a larger current drawn from the source. In the CM-TTC based NCS, a shorter average ST means energy reduction in both the MTJ branch and the CM-TTC circuit. As a result, the total energy consumption of both the typical NCS and the CM-TTC based NCS is reduced by increasing the MTJ current. However, increasing the MTJ current will reduce the energy consumption more effectively in the CM-TTC based NCS compared with the typical NCS. This is due to the fact that increasing I_{MTJ} will reduce the mean value of ST faster than the worst-case ST. As shown in Fig. 8, the energy consumption of CM-TTC based NCS decreases from 1088fJ at $I_{\text{MTJ}} = 120\mu\text{A}$ to 534fJ at $I_{\text{MTJ}} = 170\mu\text{A}$. Hence, the improvement in CM-TTC based NCS energy consumption increases from 53% at $I_{\text{MTJ}} = 120\mu\text{A}$ to 66% at $I_{\text{MTJ}} = 170\mu\text{A}$ compared with the typical NCS.

D. Optimization Delay on CM-TTC

The delay of the CM-TTC can change the total energy consumption of the CM-TTC-based NCS. The faster the MTJ current cuts off after switching (less delay), the lower the energy consumption of NCS is. On the other hand, to increase the speed of the CM-TTC, a higher current should be driven in this circuit. This leads to the higher energy consumption of the CM-TTC. Hence, an optimal delay that minimizes the total energy consumption of the CM-TTC based NCS should be calculated. The NCS and CM-TTC energy consumption are shown in Fig. 9 (a). To find the optimal delay value, the total energy consumption of the CM-TTC based NCS is calculated versus different delays of CM-TTC for different MTJ currents (i.e. $120\mu\text{A}$ - $170\mu\text{A}$) as shown in Fig. 9 (b). The CM-TTC delay is determined as the difference between the time in which the MTJ is switched and the time in which the MTJ current will

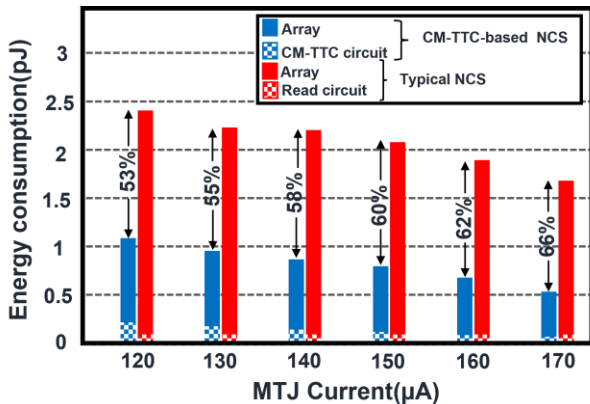


Fig. 8. The total energy consumption comparison of the CM-TTC-based NCS and the typical NCS for different MTJ currents. The improvement of the total energy consumption of the CM-TTC-based NCS in comparison with the typical NCS is shown as a percentage.

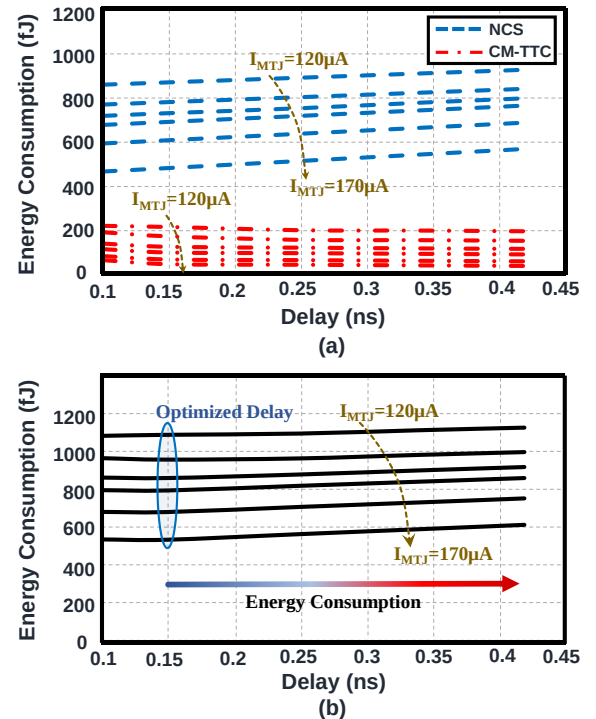


Fig. 9. The total energy consumption of the CM-TTC-based NCS, the energy consumption of the CM-TTC and energy consumption of the NCS versus delay of the CM-TTC for different MTJ currents. When the delay of the CM-TTC is 0.15ns, the total energy consumption is minimized at 534fJ for $I_{\text{MTJ}} = 170\mu\text{A}$ up to 1088fJ for $I_{\text{MTJ}} = 120\mu\text{A}$.

be cut off (TG1 turns off). The optimal delay is the delay at which the total energy consumption of NCS is minimized. Consequently, the optimal CM-TTC delay is achieved as 0.15ns. Hence, the minimum total energy at the optimal CM-TTC delay (0.15ns) changes from 1088fJ for $I_{MTJ}=120\mu A$ up to 534fJ for $I_{MTJ}=170\mu A$ as shown in Fig. 9 (b).

E. Effect of CM-TTC on the Speed Performance

As mentioned before, according to the 6-sigma worst-case analysis of ST, the minimum Pulse1 duration is to be chosen as 13.5ns and the overall delay of the read-circuit is considered as approximately 1ns [25] to guarantee the proper MTJ switching activity. Hence, the total delay of the NCS is 14.5ns (i.e. 68.9MHz). In the CM-TTC-based NCS, the overall delay decreases due to the simultaneous tracking of the MTJ state. Hence, considering the delay of CM-TTC (0.15ns) and the ST mean value of 6.97ns, the overall delay for the CM-TTC-based NCS is obtained to be 7.12ns. As a result, the maximum frequency of 140MHz at $V_{DD}=1V$ (i.e. 2X faster speed) can be achieved for the proposed system. Noted if a higher current will be applied to the MTJ, it will be switched faster. This will increase the general speed of the network at the cost of higher energy consumption. As shown in Fig.10 (a), the frequency of the CM-TTC-based NCS increases from 140MHz at $I_{MTJ}=120\mu A$ to 398.98MHz at $I_{MTJ}=170\mu A$ while the frequency of the typical NCS increases from 68.9MHz at $I_{MTJ}=120\mu A$ to 154MHz at $I_{MTJ}=170\mu A$ (Fig. 10 (a)). It shows by increasing the current the speed improvement due to using CM-TTC will increase from 2x at $I_{MTJ}=120\mu A$ to 2.6x at $I_{MTJ}=170\mu A$. To investigate the effect of MTJ scaling on overall speed, the MTJ size is changed from 70% to 110% of the size used in Table I (40nm×116nm×1.5nm) and ST and CM-TTC delay are extracted for $V_{DD}=1V$ (Fig. 10 (b)). Decreasing the MTJ size, its switching time decreases and the delay of CM-TTC will slightly increase because smaller MTJs have higher resistance that will lead to smaller I_{MTJ} that will reduce the speed of CM-TTC slightly from 0.15ns (40nm×116nm×1.5nm) to 0.25ns (0.7 of 40nm×116nm×1.5nm). On the other hand, by scaling MTJ its critical current will reduce linearly [33], leading to a linear reduction in the size of series transistors (i.e. M1 and TG1).

E. Comparison with other designs

Here, different features of the CM-TTC based NCS and its effectiveness have been compared with other reported designs in the literature. The results of the proposed CM-TTC is compared with the best previously proposed methods (RTS [11] and SDWT [13]) and summarized in Table II. Because the RTS

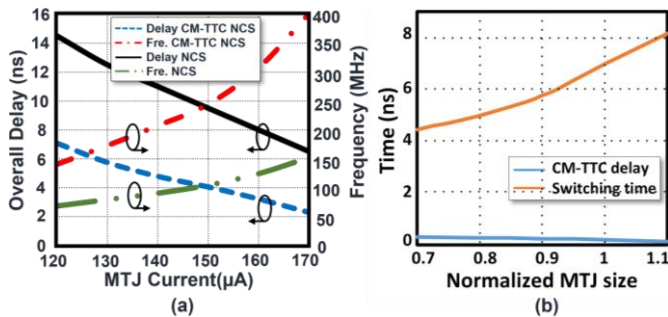


Fig. 10. (a) The overall delay/frequency of the CM-TTC-based and typical NCS versus different MTJ currents. (b) The CM-TTC delay and ST versus MTJ size normalized to the used size in Table I (40nm×116nm×1.5 nm).

circuit [11] utilizes the voltage variations ($\sim 40mV$) of PL, FL or both to detect the MTJ switching, it requires an amplifier to amplify the voltage changes. On the other hand, the amplifier in the RTS circuit is turned on during approximately 90% of the stimulating phase. This will reduce the energy efficiency of the RTS circuit. Moreover, utilizing the voltage variations for sensing the switching increases the delay of the RTS circuit.

This will also affect the energy efficiency of the RTS-based NCS since during the delay not only the RTS consumes energy but also a high current is passing through the MTJ (MTJ is in P-state with low resistance). However, in the CM-TTC-based NCS, due to the high changes in the MTJ current after switching, there is no need to use an extra amplifier. This will decrease the energy consumption of CM-TTC as well as its delay in comparison with the RTS circuit. The power consumption of CM-TTC is measured 37.8 μW that is 12.8% and 14.1% less than RTS and SDWT, respectively. Moreover, higher changes of I_{MTJ} ($\sim 40\%$) due to switching compared with V_{MTJ} changes ($\sim 15\%$) as shown in Fig. 5 will reduce the delay of CM-TTC compared with RTS and SDWT by 73% and 81%, respectively. Finally, the size of the proposed CM-TTC is estimated to be 2.7 μm^2 that is 95% and 96.4% smaller than RTS and SDWT, respectively. This large size reduction is due to the lack of capacitors in the CM-TTC structure.

Another advantage of the CM-TTC compared with state-of-the-art voltage-based methods is its higher robustness in the presence of process variations. To this end, an MC simulation with 1000 iterations is run on the CM-TTC based NCS. Then, the current comparator output voltage (V_{COM}) variations and the buffer decision point (DP) variations are calculated (Fig. 11). To consider the effect of process variations on transistors, parameters reported in [34] are used. As shown in Fig. 11 (a), in the CM-TTC-based NCS, the DP has a normal distribution with 7.13mV standard deviation while the V_{COM} distribution is

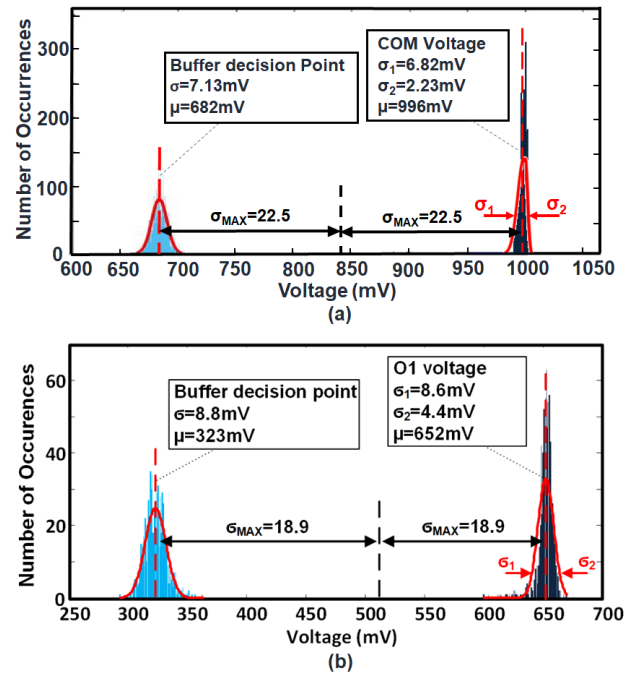


Fig. 11. The histogram and fitted Gaussian function of V_{COM} , V_{O1} and DP due to process variations for the MC simulation with 1000 iterations while using (a) CM-TTC-based NCS and (b) RTS-based NCS [11].

TABLE II

PERFORMANCE COMPARISON OF PROPOSED CM-TTC, RTS AND SDWT CIRCUITS

Parameters	Proposed CM-TTC	RTS [11]	SDWT [13]
Technology	65-nm CMOS	65-nm CMOS	65-nm CMOS
V_{DD} (V)	1	1	1
Delay (ns)	0.15	0.55	0.79
Switching Time (ns)	6.97	7.13	5.28
Power (μ W)	37.8	43.34	44.04
Area (μm^2)	2.7	55.08	74.358

negatively skewed with a right tail standard deviation (σ_2) of 2.23mV and left tail standard deviation (σ_1) of 6.82mV. It means that a 22.5σ variation on both DP of buffer and V_{COM} is needed to face a fault which means the chance of error is almost zero (Fig. 11 (a)). Fig. 11 (b) shows the histogram and fitted Gaussian function of V_{O1} and DP for the RTS circuit [11]. The probability of erroneous output of NCS (18.9σ variation on V_{O1} and DP voltages) is higher than CM-TTC.

V. CONCLUSION

In this paper, a new current-mode track and termination circuit (CM-TTC) for real-time sensing of MTJ state in the NCS application has been presented. The conventional voltage-mode sensing circuit is replaced with the proposed CM-TTC, to improve the energy efficiency and speed of NCS. In all state-of-the-art designs, the voltage changes on nodes across the MTJ have been used to detect the MTJ switching. However, the CM-TTC detects the MTJ switching by properly sensing the MTJ current and terminates its current immediately. Simulation results show around 2.7X bigger changes in the MTJ current compared with the MTJ voltage due to switching that will ease the design of the current-mode track and termination circuit. The simulation results in 65-nm CMOS technology confirm that the energy consumption of the proposed CM-TTC-based NCS and its overall speed are approximately improved by 53% and 2X in comparison with the typical NCS.

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